
Cristar

Overall Team Feedback from SMEs

Strong experimental concept with good access to equipment/support laboratories and a strong experiment design. The proposal would benefit from clearer business case, operations plan, laser description, and funding strategy.

Importance and Business Case

- Connection to Canada's space sector is thin
- The financial value of using suborbital launches to produce a relatively small quantity of crystals is uncertain
- The scope of the problem and the overall business case for space-based crystallization could be clearer
- It is unclear whether the cost of suborbital flights, plus the reduced number of crystals that can be produced compared to on Earth, is relevant for the pharmaceutical industry
- Unclear how this work would reduce the number of satellites in low Earth orbit

Goals and Experimental Design

- Goals could be more detailed and descriptive, including whether NordSpace is the only company that could produce these crystals as a business and who else in the space sector would benefit
- The hypothesis could be more detailed, for example, comparing crystals from 0-g with a laser to normal 1-g without a laser in terms of quality/size

Laser Classification and Safety

- The proposal mentions class 1 compliance for the laser, but the attached information identifies a class 3 or class 3b laser; this needs discussion with NRC
- It is not yet clear whether the use of a class 3b laser is acceptable for NRC
- Safety procedures need to ensure the laser is off and include redundancy checks before opening the Pelican case to inspect the hardware

Operations, Gravity Levels and Vibration

- Ambient vibration from the aircraft is not fully taken into account and may have a large impact on the science
- The in-flight operations plan is somewhat disorganized
- There may be limitations in testing 2-g in addition to 0-g in the same parabolas unless data collection is continuous and uses the same sample
- Operations in 2-g and during the transition to 0-g should be avoided, since it is hard to move around and control the experiment during these periods

Equipment and Procurement

- The equipment procurement plan is quite light and would benefit from more detail

Funding Strategy

- Funding strategy was described as generic, with nothing definitive beyond applications to two grants and internal support
- Some plans are in place for sponsorship, but much more specific and larger community and student funds or grants should be pursued
- The \$1000 Special Activity Grant alone would likely be spent very quickly on a small number of material orders/activities (need to also consider travel/lodging for flight campaign)